

Earth Science

Land Sciences

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Short Title: Earth Science
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author NMCCARTHY
Credits 5

Level: Introductory

Description of Module / Aims

This module is designed to give first year students of land-based courses an introduction to the earth, its origin and basic makeup, the interactions which occur between the earth and its inhabitants, its atmosphere, and artificial features of human origin. Topics covered in this course will include geology, ecology, countryside management, climatology and meteorology.

Programmes

ENVI -0001	BSc in Forestry (WD_SFORS_D)	1 / 2 / M
ENVI -0001	BSc in Forestry (WD_SFORS_D)	2 / 4 / E

Indicative Content

- An introduction to the geology of the earth, including geological time, process of rock formation, rock types, weathering and distribution, recent earth events shaping the Irish landscape of today
- Ecology and countryside management, biodiversity, ecosystems, habitats and biological niches; species interaction, competition, predation, parasitism, mutualism; succession, climax community; EU habitat directives; native flora and fauna; historical and archaeological features
- Climatology and meteorology, terms and maps; evaporation and transpiration; data recording; global climate change; events of historical significance; influence of weather and climate on forest growth and development
- Each student will undertake a field- or laboratory-based project on each of the above topics
- Field trips to river and woodland ecosystems, studying flora, fauna and geology of places such as Wicklow, Comeraghs or Killarney National Park; visit to a meteorological station

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Examine the geology of the earth: rock types, distribution, and relationship to soil formation.
2. Examine the biodiversity to be found on the earth, and discuss species interaction, native and artificial habitats, and the influence of EU policy on these.
3. Identify the effects of climate and climate change in Ireland on forests and trees.
4. Use climatology and meteorology terms and construct graphs from meteorological data and statistics, and draw conclusions from them.
5. Discuss the role of earth science and its interaction with the management of forests in the landscape.

Learning and Teaching Methods

- Lectures.
- DVD's.
- Independent learning.
- Field trips.

- Project.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Project</i>	35%	1,5
<i>Presentation</i>	35%	2,5
<i>Project</i>	30%	3,4

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Aalen, F.H.A., K. Whelan and M. Stout. *Atlas of the Irish rural landscape*. 2nd ed.. Ireland: Cork University Press, 1997.
- Carver, M. *Environmental Biology*. 1st ed.. UK: Cambridge University Press, 2009.
- Collins, J.F. and T. Cummins. *Agroclimatic Atlas of Ireland*. Ireland: AGMET Group, 1996.
- Mitchell, F. and M. Ryan. *Reading the Irish Landscape*. 3rd ed.. Ireland: Town House, 1997.

- Thompson, G.R. and J. Turk. *Earth Science and the Environment*. 4th ed.. UK: Cengage Learning Inc., 1995.
- Whelan, T. *Climate, weather and Irish agriculture*. Ireland: SOS Free Stock, 2004.

Supplementary Material

- Smith, R.L. and T.M. Smith. *Elements of ecology*. 9th ed.. UK: Pearson, 2015.

Requested Resources

- Lecture Room: Loose Seated